

	Department of Computer Science & Engineering (CSE) School of Engineering, Technology & Sciences INDEPENDENT UNIVERSITY, BANGLADESH (IUB)		
Computer Science and Engineering (CSE) Program (Undergraduate)			
COURSE OUTLINE		Summer 2026	FULL MARKS: 100
Course Code	CSE200	Course Title	Computer Programming II
Course Type	Core	Prerequisite (if any)	CSE100 + Lab
Credit Value	4 (T: 3, L: 1, T: 0)	Contact Hours / Week	3 hrs (Theory), 1.5 hrs (Lab), 1.5 hrs (Tutorial)
Year	2026	Semester	Summer
Section	7	Room	CSCLab1
Day	Wednesday	Class Timing	09:40-11:10
Instructor	Mir Sazzat Hossain	Designation	Research Assistant
Email	sazzat@iub.edu.bd	Office hours	By appointment via email
COURSE DESCRIPTION			
The main objective of this course is to develop a strong foundation of computer programming knowledge within the CSE students for real-life problem solving from the perspective of a software engineer. Students will gain mastery of the concepts of both procedural language and object-oriented (OOP) paradigm. The advance knowledge of control statements (e.g., conditional statements, loops), program modularization (e.g., functions, class, and methods), data storage, management, and manipulation (e.g., arrays, list, sets, tuple, dictionary), and secondary storage management (e.g., File system) are covered in detail. The concept of object-oriented programming, e.g., class, object, methods, inheritance, polymorphism, encapsulation are introduced and used to solve problems. Additionally, error handling mechanisms (e.g., exception handling) are also discussed. Each lecture would involve solving several programming problems on the learned topic using designated IDE. After successful completion of the course, a student will possess the mastery of breaking down a complex programming problem into smaller parts, solve them and write the solution in python.			
COURSE OBJECTIVE			
a. Advance skill on the understanding and use of structured programming concepts in solving real-life computational problems as sketched above. b. Detailed understanding of the OOP paradigm and developing skills to use them in solving real-life computational problems.			

	c. Learn in detail the memory management (both primary and secondary), data structure and error handling mechanism. d. Grow in confidence with programming skills in dealing with non-trivial problem solving.
	COURSE POLICY
	It is the student's responsibility to gather information about the assignments and cover topics if they miss lectures. Regular class attendance is mandatory. Following policy as per the faculty manual will be adopted: Points will be taken off for missing class tests. Without 70% of attendance, sitting for the final exam is NOT allowed. According to the IUB policy, students must enter the classroom within the first 10 minutes to get the attendance submitted. a. The date and syllabus of the lectures, quizzes, midterm, and final exam are already given here, however, announcements will be given ahead of time. There is NO provision for make-up quizzes. b. The reading materials for each class may be given before that class so that students may have a cursory look into the materials. All materials (lecture notes, supporting reading materials, problem sheets and the solutions) will be made available through google classroom (classroom.google.com). c. Class participation is vital for a better understanding of the subject matter. Class will be conducted in an interactive environment where both the teacher and the students must pose questions and discuss solutions for better understanding. d. Students must attend the scheduled tutorial classes of the Research Assistants. Attending and performing in the tutorial session is mandatory and is part of the overall course evaluations. e. Students must maintain the IUB code of conduct and ethical guidelines offered by the school of engineering and computer sciences.
	ACADEMIC DISHONESTY
	a. A student who cheats, plagiarizes, or furnishes false, misleading information in the course is subject to disciplinary action up to and including an F grade in the course and/or suspension/expulsion from the University. b. Students must maintain the code of IUB. c. The goal of homework is to give you practice in mastering the course material. Consequently, you are encouraged to collaborate on problem sets. In fact, students who form study groups generally do better on exams than do students who work alone. If you do work in a study group, however, you owe it to yourself and your group to be prepared for your study group meeting. Specifically, you should spend 30-45 minutes on each problem beforehand, trying to solve it yourself. If your group is unable to solve a problem, talk to other groups or ask your recitation instructor or the teaching assistant assigned to your class. d. You must write up each problem solution by yourself without assistance. It is a violation of this policy to submit a problem solution that you cannot orally explain to a member of the course staff. e. No collaboration whatsoever is permitted during the examination. f. Plagiarism and other anti-intellectual behavior cannot be tolerated in any academic environment that prides itself on individual accomplishment. If you have any questions about the collaboration policy or feel you may have violated it, please talk to one of the course staff. Although the course staff is obligated to deal with cheating appropriately, we are more understanding and lenient if we learn of the transgression from the transgressor themselves rather than from a third party or from ourselves.

STUDENT WITH DISABILITIES AND STRESS

Students with disabilities are required to inform the Department of Computer Science & Engineering of any specific requirements for classes or examinations as soon as possible. Additionally, if you experience significant stress or worry, changes in mood, or problems eating or sleeping this semester, whether because of this or any other courses or factors, please do not hesitate to reach out immediately, at any hour, to any of the course's heads to discuss.

NON DISCRIMINATION POLICY

The course and University policy prohibit discrimination based on race, color, religion, sex, marital or parental status, national origin or ancestry, age, mental or physical disability, sexual orientation, or military status. If you see either the course instructor or any other person related to the course showing any form of discrimination, please inform the proctor's office of the wrongdoing.

COURSE CONTENT

- a. Conditional Statements (Nested)** **1.5 Week (4.5h Th + 1.5h L + 1.5 Tu)**
Use of conditional statements in solving complex decision-making problems. Learn the use of nested conditional statements and associated technical details: if-else, if-elif-else, nested if.
- b. Loops (Nested)** **1.5 Week (4.5h Th + 1.5h L + 1.5 Tu)**
Solving complex recurring problems using loops. Learn in detail: while loop, for loop, nested loop, break statement, continue statement, pass statement.
- c. Functions – calls, parameters, scope of variables** **1.5 Week (4.5h Th + 1.5h L + 1.5 Tu)**
Introduce the concept of functions with function calls, use of parameters while calling functions and realization of the variables scope in the context of functions
- d. Recursion** **1 Week (3h Th + 1.5h L + 1.5 Tu)**
Detailed understanding of the recursive function calls with its behavior and realizing it with a physical world example.
- e. Data structure (List, Tuple, Set, Dictionary)** **3.5 Weeks (10.5h Th + 6h L + 6h Tu)**
Detailed understanding on the data storage and management and its importance from real life problem scenarios. Define, store, and manipulate List, Tuple, Set and Dictionary for data management. Learn in detail associated library methods and techniques. Multidimensional data structures, e.g., 2D list should be learnt.
- f. Modularization (Function) and Introduction of a Library** **1 Weeks (3h Th + 3h L + 3 Tu)**
Define function from the perspective of modularization / reusability / maintainability / separation of concept / divide-and-conquer in solving complex and recurring problems, Use of function: Argue from real life usage scenario (e.g., an automated chocolate factory), Show difference between code solution with and without function, Allow student to solve problems (conceptually) using functions, Structure of a function, e.g., parameters, returns value, calling a function, return from a function, With simple example show step-by-step how a function executes. Let the student understand how a program execution flows when a function is called, executed, and returned, Example problems and their implementation using functions.
- g. Files (File I/O)** **0.5 Week (1.5 Th + 1.5h L)**
Introduce the concept of File as a secondary storage structure / persistence storage, what is the distinction between in memory storage (e.g., variables / data structures) and File, learn how to create

LECTURE LESSON PLAN WITH MAPPING OF CLO, TEACHING, AND ASSESSMENT STRATEGIES

Week	Topic	Teaching-Learning Strategy	Assessment Strategy	CLOs
1	Conditional Statements (Nested)	Lecture (4.5h), Lab (1.5h), Tutorial (1.5h)	Class Test/Mid term	
2, 3	Loops (Nested)	Lecture (4.5h), Lab (1.5h), Tutorial (1.5h)	Class Test / Mid Term / Lab work Progressive project	CLO1
3, 4	Functions - calls, parameters, scope of variables, generators	Lecture (4.5h), Lab (1.5h), Tutorial (1.5h)	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
5, 6	Built-in data types - List (+2D)	Lecture (4.5h), Lab (3h), Tutorial (3h)	Class Test / Mid Term / Lab work Progressive project	CLO1 CLO2
Mid Week				
7	Recursion	Lecture (3h), Lab (1.5h), Tutorial (1.5h)	Class Test / Final Term / Lab work Progressive project	CLO1 CLO2
8	Built-in types - List, Tuple, Set, Dictionary	Lecture (3h), Lab (1.5h), Tutorial (1.5h)	Class Test / Final Term / Lab work Progressive project	CLO1 CLO2
9	Built-in types - List, Tuple, Set, Dictionary	Lecture (3h), Lab (1.5h), Tutorial (1.5h)	Class Test / Final Term / Lab work Progressive project	CLO1 CLO2
10	Modularization (functions) and introduction of a library - pygame	Lecture (3h), Lab (3h), Tutorial (3h)	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3
11	Exception Handling - try-catch	Lecture (3h), Lab (1.5h), Tutorial (1.5h)	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3

12	Files (File I/O) - text mode	Lecture (3h), Lab (1.5h)	Class Test / Final Term / Lab work Progressive project	CLO2 CLO3
12, 13	Python - Class, Object	Lecture (4.5h), Lab (1.5h), Tutorial (3h)	Class Test / Final Term / Lab work Progressive project	CLO3
14	Review	Lecture (3h), Lab (1.5h)		

ASSESSMENT AND EVALUATION						
Assessment Type	Assessment Tools	Marks Distribution	Assessment No	CLO Assessed	Blooms Category	Sub Total
Continuous Assessment	Class Tests	15%	Mid Test - 1			50%
			Mid Test - 2			
			Final Test - 3			
			Final Test - 4			
	Lab Assessment	10%		CLO3	Determine, Apply, Develop	
	Progressive Project	15%				
	Tutorial	10%				
Summative Assessment	Mid Term Examination	20%		CLO1	Determine, Apply	50%
	Final Term Examination	30%		CLO2	Analyze, Determine, Apply, Develop	
Total						100%

CLO 1 Evaluation Rubric		Blooms Level: Analyze
Evaluation Category	Evaluation Definition	Marks Allocation Max – 25 CLO pass – 60%
Program Compile & Run	Build and run program using IDE, show desired output in the console window	Max – 5 Range 0 - 5
Logical Operations	Apply logical operations to connect two or more expressions. Use logical operator such as AND, OR, NOT and so on.	Max – 5 Range 0 - 5
Correctness	Make the program error free, mathematically, and logically make it accurate	Max – 5 Range 0 - 5
Completeness	Complete the project with implementing all the given requirements	Max – 5 Range 0 - 5
Coding Standard	Improve readability and maintainability of the code, write code in a way to make it easy for reuse, to figure out mistakes	Max – 5 Range 0 - 5
CLO 2 Evaluation Rubric		Blooms Level: Determine, Apply
Evaluation Category	Evaluation Definition	Marks Allocation Max – 25 CLO pass – 60%
Solution Design	Conceptual design of the proposed solution.	Max – 5 Range 0 - 5
Design Correctness	Explain relevance of the solution in relation to the stated problem.	Max – 5 Range 0 - 5
Implementation	Correctness of the developed program – output, coding standard and others.	Max – 5 Range 0 - 5
Completeness	Comprehensiveness of the program in relation to problem definition – does it meets all the requirements?	Max – 5 Range 0 - 5

Explain	Able to defend / reason on the solution they did in relation to the problem definition, e.g., why a specific approach is used not the other?	Max – 5 Range 0 - 5
CLO 3 Evaluation Rubric		Blooms Level: Develop
Evaluation Category	Evaluation Definition	Marks Allocation Max – 25 CLO pass – 60%
Solution Design	Conceptual design of the proposed solution.	Max – 5 Range 0 - 5
Design Correctness	Explain relevance of the solution in relation to the stated problem.	Max – 5 Range 0 - 5
Implementation	Correctness of the developed program – output, coding standard and others.	Max – 5 Range 0 - 5
Completeness	Comprehensiveness of the program in relation to problem definition – does it meets all the requirements?	Max – 5 Range 0 - 5
Explain	Able to defend / reason on the solution they did in relation to the problem definition, e.g., why a specific approach is used not the other?	Max – 5 Range 0 - 5

REFERENCE BOOK AND ADDITIONAL MATERIALS	
	a. Python Crash Course, 2nd Edition: A Hands-On, Project-Based Introduction to Programming, Eric Matthes b. Learning Python, 5th Edition, Mark Lutz c. Automate the Boring Stuff with Python, 2nd Edition: Practical Programming for Total Beginners, Al Sweigart

VERIFICATION	
Prepared by: Dr. Razib Hayat Khan <i>Course Coordinator</i> Date: 14/May/2026	Moderated by: _____ Lead <i>OBE Committee for CSE</i> Date:..... Checked by: <i>Head, Dept of CSE, IUB.</i> Date:.....
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